

6

Table 2 shows how a bird called the bluethroat (*Luscinia svecica*) is classified by biologists.

Table 2

Taxon	Name of taxon
Domain	Eukaryota
	Animalia
	Chordata
	Aves
	Passeriformes
	Muscicapidae
Genus	
Species	

0 6 . **1** Complete **Table 2** by filling the seven blank spaces with the correct terms.

[2 marks]

A group of scientists investigated genetic diversity in different species of bird. For each species, the scientists:

- collected feathers from a large number of birds
- extracted DNA from cells attached to each feather
- analysed the samples of DNA to find genetic diversity.

Table 3 summarises their results.

Table 3

Species of bird	Number of genes examined	Number of genes examined that showed genetic diversity
Willow flycatcher	708	197
House finch	269	80
Bluethroat	232	81

0 6 . 2 In this investigation, what is meant by **genetic diversity**?

[1 mark]

0 6 . 3 The scientists concluded that the bluethroat showed greater genetic diversity than the willow flycatcher. Explain why they reached this conclusion. Use calculations to support your answer.

[2 marks]

Turn over for the next question

